Mahatma Education Society's

Pillai College of Arts, Commerce & Science (Autonomous)

Affiliated to University of Mumbai

'NAAC Accredited 'A' grade (3 cycles)'
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ISO 9001:2015 Certified

MAHATMA EDUCATION SOCIETY'S

PILLAI COLLEGE OF ARTS, COMMERCE & SCIENCE

(Autonomous)

NEW PANVEL

PROJECT REPORT ON

"Zero Waste Pantry Manager"

IN PARTIAL FULFILLMENT OF

Degree of Bachelor Of Science in Computer Science SEMESTER IV :-2025-2026

PROJECT GUIDE : PROF. Shubhangi Pawar Mam

SUBMITTED BY: Gopal Ingale

ROLL NO: 5296

DIVISION :- SYCS / A

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Certificate

This is to Certify that MR. **Gopal Ingale** of **S.Y.B.Sc. CS Semester IV** has Completed the Practical work in the Subject of the **Hybrid DBMS** During the academic year **2025-2026** Under the Guidance of **Prof. Shubhangi Pawar Mam** Being the partial requirement for the Curriculum of **Degree Of Bachelor Of Science in Computer Science , University Of Mumbai**

Place:

Date:

Name & Signature of Faculty:

Name & Signature of External:

Introduction

The **ZeroWastePantry Management System** is a web-based application designed to help reduce food wastage by enabling individuals to donate surplus food to designated collection centers. In many households, restaurants, and events, a significant amount of food is wasted due to lack of proper management or distribution channels. At the same time, many people in society suffer from food shortages. This project aims to bridge the gap between food donors and distribution centers by providing a centralized digital platform where users can easily donate food and administrators can manage the collection and

distribution process efficiently.

The system is developed using **JSP (Java Server Pages), Servlets, HTML, CSS, and MySQL database**. It follows a structured **MVC (Model–View–Controller) architecture**, where the presentation layer is handled through JSP pages, the business logic is controlled through Servlets, and the database interactions are managed through DAO classes and SQL queries. This architecture helps maintain separation of concerns and makes the application easier to maintain and expand.

The application allows users to register, log in, and donate food items by specifying details such as food type, quantity, and location. These donations are stored in the database and can be viewed by the administrator through an admin dashboard. The administrator can monitor recent donations, view all donations made by users, and analyze donation patterns through a graphical summary.

The project also includes database-level programming concepts such as **stored procedures, triggers, conditional statements, joins, and aggregate functions** to ensure data validation and efficient data management. For example, triggers are used to validate donation inputs before insertion, while stored procedures handle donation transactions securely.

By implementing this system, the project demonstrates how technology can be used to address social issues like food wastage and hunger. The ZeroWastePantry Management System provides a simple, user-friendly, and scalable solution that encourages responsible food donation and promotes sustainable resource management.

Objective of the Project

The primary objective of the **ZeroWastePantry Management System** is to develop a web based platform that facilitates food donation and helps minimize food wastage by connecting donors with collection centers. The system aims to create an organized and efficient method for managing food donations while ensuring proper monitoring and control through an administrative interface.

One of the key objectives of this project is to allow users to easily register and log in to the system and donate surplus food items through an online interface. By providing a simple

and accessible platform, the system encourages individuals and organizations to contribute leftover food that would otherwise be wasted.

Another important objective is to provide an **admin dashboard** that allows administrators to monitor and manage all donations. The admin can view recent donations, track all donation records, and analyze donation trends through visual summaries. This helps in maintaining transparency and improving the overall management of donated food.

The project also aims to implement secure and efficient data management using **MySQL database and server-side programming with JSP and Servlets**. Data validation, triggers, and stored procedures are used to ensure that only valid and meaningful data is stored in the database.

Additionally, the system aims to demonstrate the practical implementation of web development concepts such as session management, role-based authentication, database connectivity, and dynamic content generation. By integrating these technologies, the project provides a complete solution that showcases both technical knowledge and social impact.

Overall, the main objective of this project is to design and implement a system that promotes responsible food donation, reduces food waste, and provides an efficient management platform for monitoring and distributing donated food resources.

System Requirements

The **ZeroWastePantry Management System** requires both hardware and software components to function effectively. These requirements ensure that the system runs smoothly and provides reliable performance for both users and administrators.

Hardware Requirements

The hardware requirements for this project are minimal, as the system is designed to run on a standard computer environment. The application can be developed and executed on a personal computer with basic specifications.

A computer system with a **minimum of 4 GB RAM** is recommended to ensure smooth operation of the development tools and server environment. A processor such as **Intel Core i3 or higher** is sufficient for running the application and database server. The system should also have adequate storage space, preferably **10 GB or more**, to store project files, database records, and development tools.

A stable internet connection is also beneficial for downloading required libraries, server tools, and updates during development. The system should also include standard input and output devices such as a keyboard, mouse, and display monitor.

Software Requirements

The software requirements for the project include development tools, programming languages, and database systems used to build and run the application.

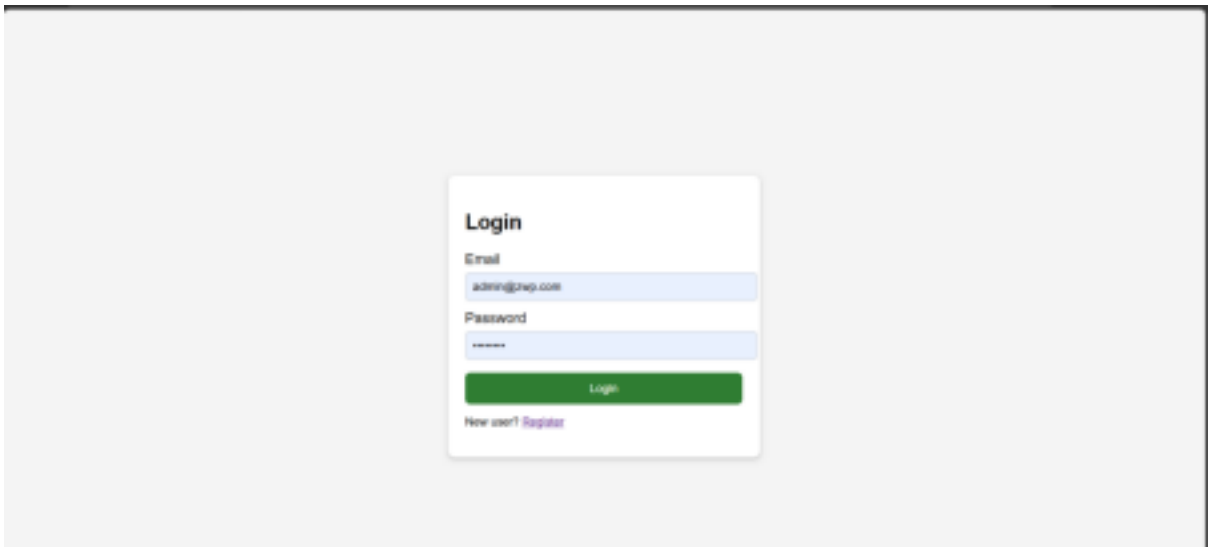
The project is developed using **NetBeans IDE**, which provides an integrated development environment for building Java-based web applications. The application uses **Java programming language** along with **JSP (Java Server Pages) and Servlets** for implementing server-side logic and dynamic web pages.
For the database management system, **MySQL** is used to store user data, donation records, and location information. The MySQL database is managed through **phpMyAdmin**, which provides a graphical interface for executing SQL queries and managing database tables.

The project is executed on a **GlassFish Server**, which acts as the application server responsible for running JSP pages and Servlets. The front-end of the application is designed using **HTML and CSS**, which provide the structure and styling for the web interface.

Additionally, the system uses **JDBC (Java Database Connectivity)** for establishing communication between the Java application and the MySQL database. This allows the application to perform operations such as inserting donation records, retrieving user information, and displaying data on dashboards.

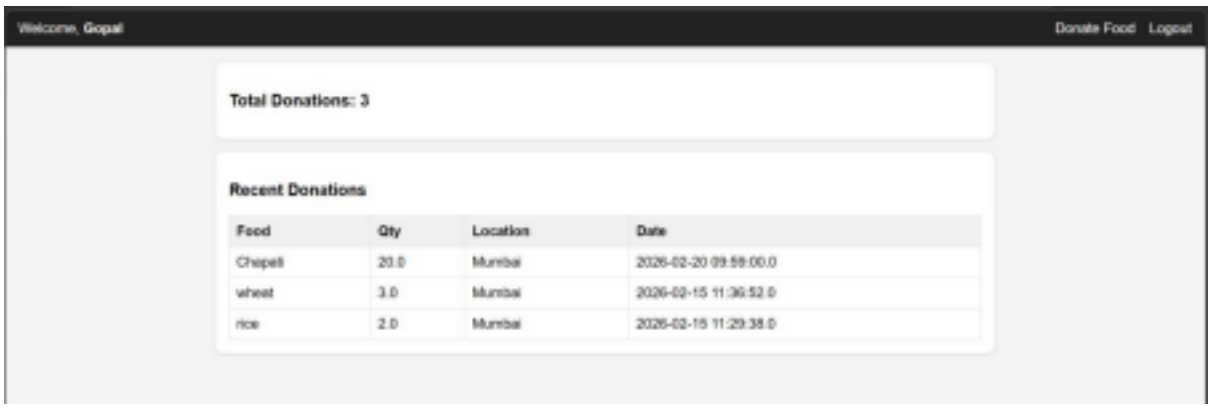
By combining these hardware and software components, the ZeroWastePantry Management System ensures efficient operation, reliable data management, and a user-friendly web interface for both donors and administrators.

Front end Design

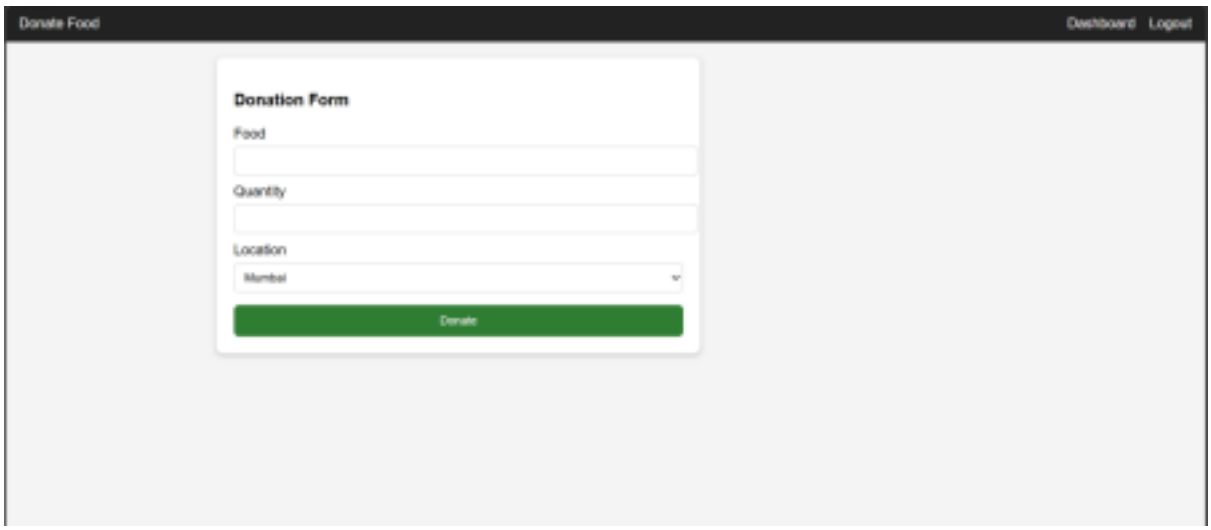


ADMIN DASHBOARD:-

User dashboard :-



Donation Page :-



Back End and PL/SQL Concepts :-

1. **Stored Procedure (Insert Donation safely)**

Use: Insert donation + set donated date automatically + validate

inputs. DELIMITER $$

```
CREATE PROCEDURE sp_add_donation(
 IN p_user_id INT,
 IN p_center_id INT,
 IN p_item_name VARCHAR(255),
 IN p_quantity VARCHAR(50)
)
BEGIN
 IF p_user_id IS NULL OR p_center_id IS NULL OR p_item_name IS NULL OR p_quantity IS
NULL THEN
 SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'All fields are required';
END IF;

 INSERT INTO donations(user_id, collection_center_id, item_name, quantity, donated_on)
VALUES(p_user_id, p_center_id, p_item_name, p_quantity, CURDATE()); END$$

DELIMITER ;
```

Call from SQL:

CALL sp_add_donation(4, 1, 'Rice', '2kg');

2) Function (Total donations of a user)

DELIMITER $$

```
CREATE FUNCTION fn_total_donations(p_user_id INT)
RETURNS INT
DETERMINISTIC
BEGIN
 DECLARE total INT;
 SELECT COUNT(*) INTO total
 FROM donations
 WHERE user_id = p_user_id;

 RETURN total;
END$$
DELIMITER ;
```

Use:

```
SELECT fn_total_donations(4);
```

3) Trigger (Validate donation before insert)

Use: Stop invalid donation like empty item name or quantity.

```
DELIMITER $$

CREATE TRIGGER trg_donation_validate
BEFORE INSERT ON donations
FOR EACH ROW
BEGIN
 IF NEW.item_name IS NULL OR TRIM(NEW.item_name) = '' THEN  SIGNAL
SQLSTATE '45000' SET MESSAGE_TEXT = 'Item name cannot be empty';  END IF;

 IF NEW.quantity IS NULL OR TRIM(NEW.quantity) = '' THEN
 SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Quantity cannot be empty';
END IF;

 IF NEW.donated_on IS NULL THEN
 SET NEW.donated_on = CURDATE();
 END IF;
END$$

DELIMITER ;
```

4) Trigger (Audit log after insert)

Use: Admin can see “who donated what” history.

Create audit table:

```
CREATE TABLE donation_audit (
 id INT AUTO_INCREMENT PRIMARY KEY,
 donation_id INT,
 user_id INT,
 action VARCHAR(20),
 action_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

Trigger:

```
DELIMITER $$

CREATE TRIGGER trg_donation_audit
AFTER INSERT ON donations
FOR EACH ROW
BEGIN
 INSERT INTO donation_audit(donation_id, user_id, action)
VALUES(NEW.id, NEW.user_id, 'INSERT');
END$$

DELIMITER ;
```

5) Cursor + Loop (Generate weekly donation summary)

Use: Graph/summary box in admin dashboard.

Create summary table:

```
CREATE TABLE donation_weekly_summary (
 week_label VARCHAR(20) PRIMARY KEY,
 donation_count INT NOT NULL
);
```

Procedure using cursor:

```
DELIMITER $$

CREATE PROCEDURE sp_build_weekly_summary()
BEGIN
 DECLARE done INT DEFAULT 0;
 DECLARE wk VARCHAR(20);
 DECLARE cnt INT;
```

```
 DECLARE cur CURSOR FOR
 SELECT CONCAT('Y', YEAR(donated_on), '-W', WEEK(donated_on)) AS wk,
COUNT(*) AS cnt
 FROM donations
 GROUP BY YEAR(donated_on), WEEK(donated_on)
 ORDER BY YEAR(donated_on) DESC, WEEK(donated_on) DESC
LIMIT 5;

 DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;

DELETE FROM donation_weekly_summary;

 OPEN cur;
 read_loop: LOOP
 FETCH cur INTO wk, cnt;
 IF done = 1 THEN
 LEAVE read_loop;
 END IF;

  INSERT INTO donation_weekly_summary(week_label, donation_count)
VALUES(wk, cnt);
 END LOOP;

 CLOSE cur;
END$$

DELIMITER ;
```

Run:

```
CALL sp_build_weekly_summary();
SELECT * FROM donation_weekly_summary;
```

6) Exception Handling (Handlers) in procedure

Use: If something fails inside procedure, rollback.

```
DELIMITER $$

CREATE PROCEDURE sp_safe_donation(
 IN p_user_id INT,
 IN p_center_id INT,
 IN p_item_name VARCHAR(255),
```

```
 IN p_quantity VARCHAR(50)
)
BEGIN
 DECLARE EXIT HANDLER FOR SQLEXCEPTION
 BEGIN
 ROLLBACK;
 SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Donation failed, rolled back';
END;

 START TRANSACTION;

 INSERT INTO donations(user_id, collection_center_id, item_name, quantity, donated_on)
VALUES(p_user_id, p_center_id, p_item_name, p_quantity, CURDATE());
 COMMIT;
END$$

DELIMITER ;
```

7) Database Event Scheduler (Auto-run summary daily)

Use: Auto update admin dashboard summary every day.

Enable scheduler (once):

```
SET GLOBAL event_scheduler = ON;
```

Create event:

```
DELIMITER $$

CREATE EVENT ev_refresh_weekly_summary
ON SCHEDULE EVERY 1 DAY
DO
BEGIN
 CALL sp_build_weekly_summary();
END$$

DELIMITER ;
```

1. Stored Procedure

2. Stored Function

3. Trigger (Validation)

4. Trigger (Audit logging)

5. Cursor + Loop

6. Exception Handling (Handlers)

7. Transaction + Event Scheduler

Conclusion

The ZeroWastePantry Management System successfully demonstrates how web-based technology can be used to address the important issue of food wastage. The project provides a simple and effective platform where users can donate surplus food and administrators can monitor and manage donation activities efficiently. By connecting food donors with collection centers, the system helps reduce unnecessary food waste while promoting responsible and sustainable resource management.

The application was developed using modern web development technologies such as JSP, Servlets, HTML, CSS, and MySQL, following a structured approach to separate presentation, logic, and database layers. The system allows users to register, log in, and donate food items easily through an intuitive interface. At the same time, the admin dashboard provides the ability to view recent donations, access complete donation records, and analyze donation trends through visual summaries.

The project also implements several important database concepts such as stored procedures, triggers, joins, and aggregate functions, which ensure data validation, security, and efficient data handling. These features help maintain the integrity of the system and demonstrate the practical use of database programming concepts in a real-world application.

Overall, the ZeroWastePantry Management System highlights how technology can be utilized to support social welfare by reducing food wastage and encouraging food donation. The system provides a foundation that can be further expanded with additional features such as real-time tracking, mobile application integration, and automated notifications. With further development, this system has the potential to become a valuable tool for communities and organizations working toward sustainable food management.